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ཁྲི།

College of Science and Technology
Rinchending: Bhutan

WEB101
WEB APPLICATION FUNDAMEN
(SS2026)

Practical 1 Report

Submitted By;
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Student No : 02240369
Programme: BE 1SWE
Date: 14.03.2026

RUB Wheel of Academic Law: Academic Dishonesty

Section H2 of the Royal University of Bhutan's *Wheel of Academic Law* provides the following definition of academic dishonesty:

Academic dishonesty may be defined as any attempt by a student to gain an unfair advantage in any assessment. It may be demonstrated by one of the following:

1. **Collusion:** the representation of a piece of unauthorised group work as the work of a single candidate.
2. **Commissioning:** submitting an assignment done by another person as the student's own work.
3. **Duplication**: the inclusion in coursework of material identical or substantially similar to material which has already been submitted for any other assessment within the University.
4. **False declaration**: making a false declaration to receive special consideration by an Examination Board or to obtain extensions to deadlines or exemption from work.
5. **Falsification of data**: presentation of data in laboratory reports, projects, etc., based on work purported to have been carried out by the student, which has been invented, altered or copied by the student.
6. **Plagiarism**: the unacknowledged use of another's work as if it were one's own.

Examples are:

- verbatim copying of another's work without acknowledgement.
- paraphrasing of another's work by simply changing a few words or altering the order of presentation, without acknowledgement.
- ideas or intellectual data in any form presented as one's own without acknowledging the source(s).
- making significant use of unattributed digital images such as graphs, tables, photographs, etc., taken from test books, articles, films, plays, handouts, the internet, or any other source, whether published or unpublished.
- submission of a piece of work which has previously been assessed for a different award or module or at a different institution as if it were new work.
- use of any material without prior permission of copyright from the appropriate authority or owner of the materials used".

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ACKNOWLEDGEMENT

First of all, I would like to thank the College of Science and Technology for giving me the important required resources for me to be able to do this practical. I am also very much thankful to our WEB module teacher, Mr. Prakash Chhetri, for helping us with his guidance and support while doing the assignment, where it helped me to understand better about the basics of web application development, not just rely on ai.

AIM AND OBJECTIVES

Aim

Is to make a TikTok clone web application by using a few things like Next.js and Tailwind CSS, which are used in video reels/feeds, navigation features in the sidebar, pagination of pages, and mainly login and sign up valid forms with proper authentication.

Objectives

- Set the Next.js project using the correct configurations

- Use Tailwind CSS to style the applications

- Make a proper, clean folder structure

- Make a layout just like TikTok style, with permanently navigable sidebars

- Also creating elements like VideoCard and VideoFeed that can be used again and again

- Have many different pages on the left side of the web page, like home, following, explore, live, upload, and profile

- Mainly a standard validation form for the login and signup by using React Hook Form

Background Information

These days, new web applications are made using different frameworks and tools to make the process faster, cleaner, and easier to handle over long durations, whereas in this particular assignment the big three techs used were Next.js, Tailwind CSS, and React Hook Form.

Next.js

The best web application building framework on top of React is the best for me, as it uses JavaScript to build the interactions or make it dynamic for the users. It enables us to be able to make smaller sections of the web page, which can be used again and again, called the components. Next.js makes extra functions on top of the react, like routing of files. This way, the folders inside the app folder are made into pages in the webpage, which makes moving between the pages is easy and simple.

Tailwind CSS

The first class framework that makes styling of web pages inside the files using class names look like child's play, files such as HTML or JSX, rather than writing style codes for each line, use of Tailwind classes is able to be used directly to the components. This way the styling of the web application is much more faster and efficiently done for the whole web application.

React Hook Form

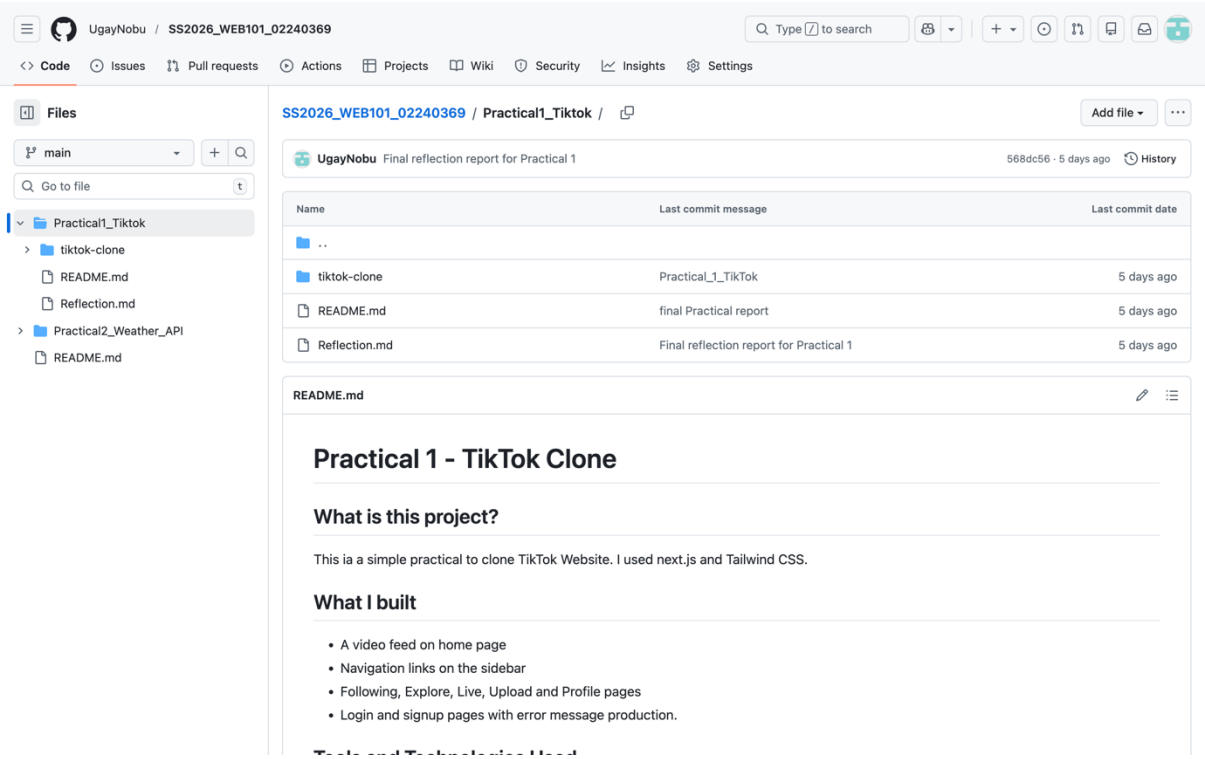
Is a very important dependence that makes handling legit forms with proper standard validation forms in the React applications. It gives easy functions to sign up, submit forms, and watch, that is making the inputs of the user to be to easir to collect with valid formats by setting rules when break of rule is broken, leading to pop up of error messages, such as when the inputs from the user is wrong. It was used in this assignment for the login and signup forms.

These three main techs will allow us to create a good structure with attractive styling and workable TikTok functions or features in the web page.

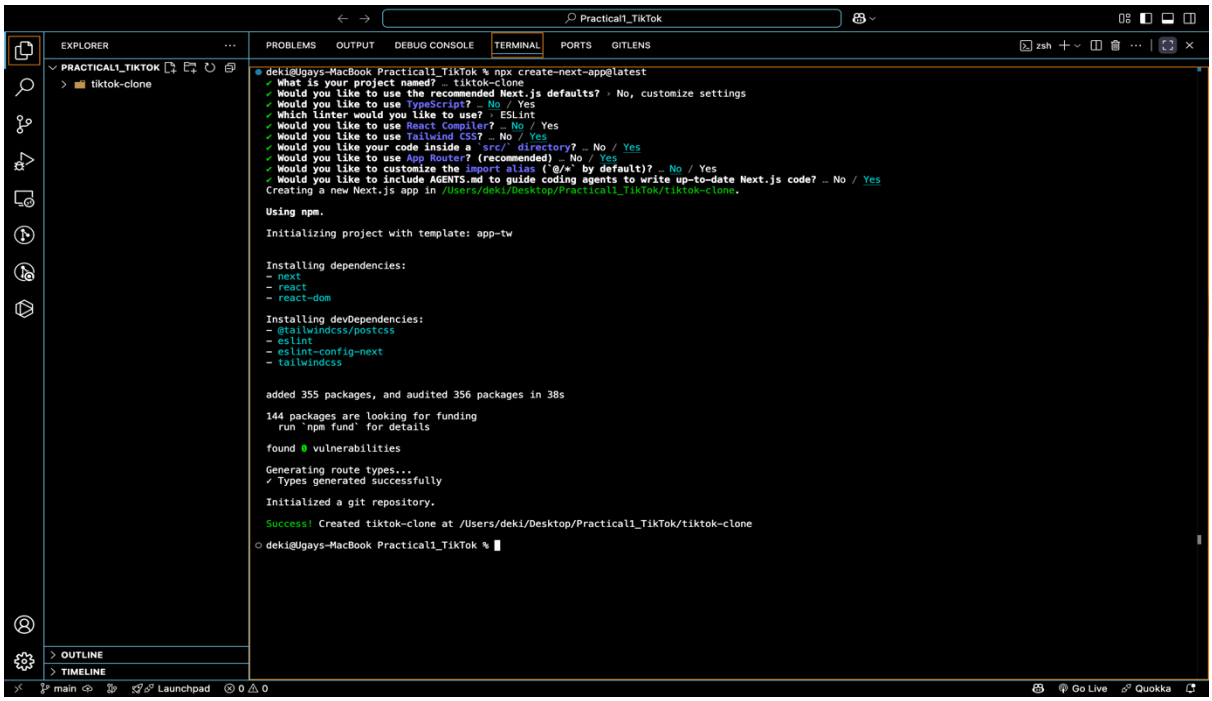
Implementation/ PROCEDURE

Part 1 - Getting Started

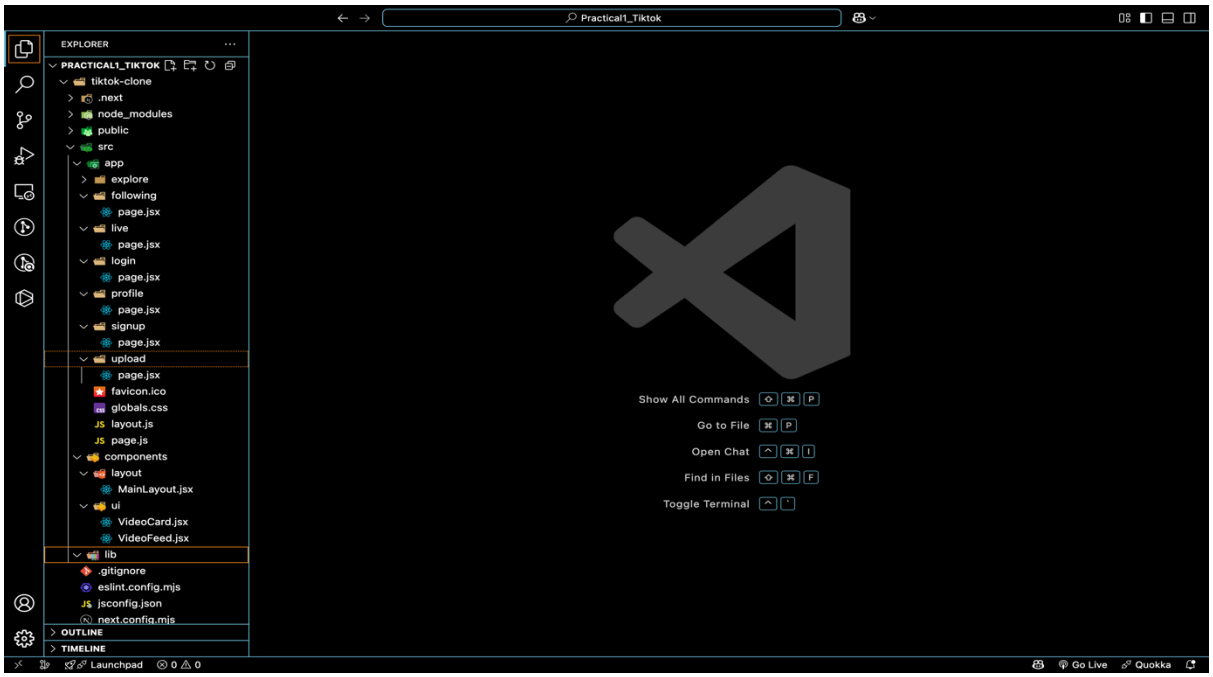
Step 1: Create a GitHub Repository and clone it to the local machine.



Step 2: Create a new Next.js project using: npx create-next-app@latest



Step 3: Clean up and folder structure



Step 4: Set up the project folder structure using terminal commands.

Step 5: Create the MainLayout.jsx layout component with navigation links.

Step 6: Update layout.js to include the MainLayout component.

Step 7: Create basic pages for Profile and Upload.

Step 8: Start the development server.

```
deki@Ugays-MacBook Practical1_Tiktok % cd tiktok-clone
deki@Ugays-MacBook tiktok-clone % npm run dev

 > tiktok-clone@0.1.0 dev
 > next dev

 ⚠ Warning: Next.js inferred your workspace root, but it may not be correct.
  We detected multiple lockfiles and selected the directory of /Users/deki/package-lock.json as the root directory.
  To silence this warning, set `turbopack.root` in your Next.js config, or consider removing one of the lockfiles if it's not needed.
    See https://nextjs.org/docs/app/api-reference/config/next-config-js/turbopack#root-directory for more information.
  Detected additional lockfiles:
    * /Users/deki/Desktop/WEB/SS2026_WEB101_02240369/Practical1_Tiktok/tiktok-clone/package-lock.json

 ▲ Next.js 16.1.6 (Turbopack)
 - Local:         http://localhost:3000
 - Network:       http://10.2.26.137:3000

 ✓ Starting...
 ✓ Ready in 686ms
  GET / 200 in 788ms (compile: 530ms, render: 258ms)
  GET / 200 in 29ms (compile: 5ms, render: 23ms)
  GET / 200 in 23ms (compile: 4ms, render: 20ms)
  GET / 200 in 24ms (compile: 4ms, render: 20ms)
  GET / 200 in 29ms (compile: 6ms, render: 23ms)
  GET / 200 in 21ms (compile: 3ms, render: 17ms)
  GET / 200 in 21ms (compile: 3ms, render: 18ms)
  GET / 200 in 23ms (compile: 4ms, render: 19ms)
  GET / 200 in 30ms (compile: 9ms, render: 20ms)
  GET / 200 in 24ms (compile: 5ms, render: 19ms)
  GET /following 200 in 62ms (compile: 36ms, render: 25ms)
  GET / 200 in 20ms (compile: 3ms, render: 16ms)
  GET /following 200 in 24ms (compile: 3ms, render: 21ms)
```

Part 2 - Web Layout and Main Interface

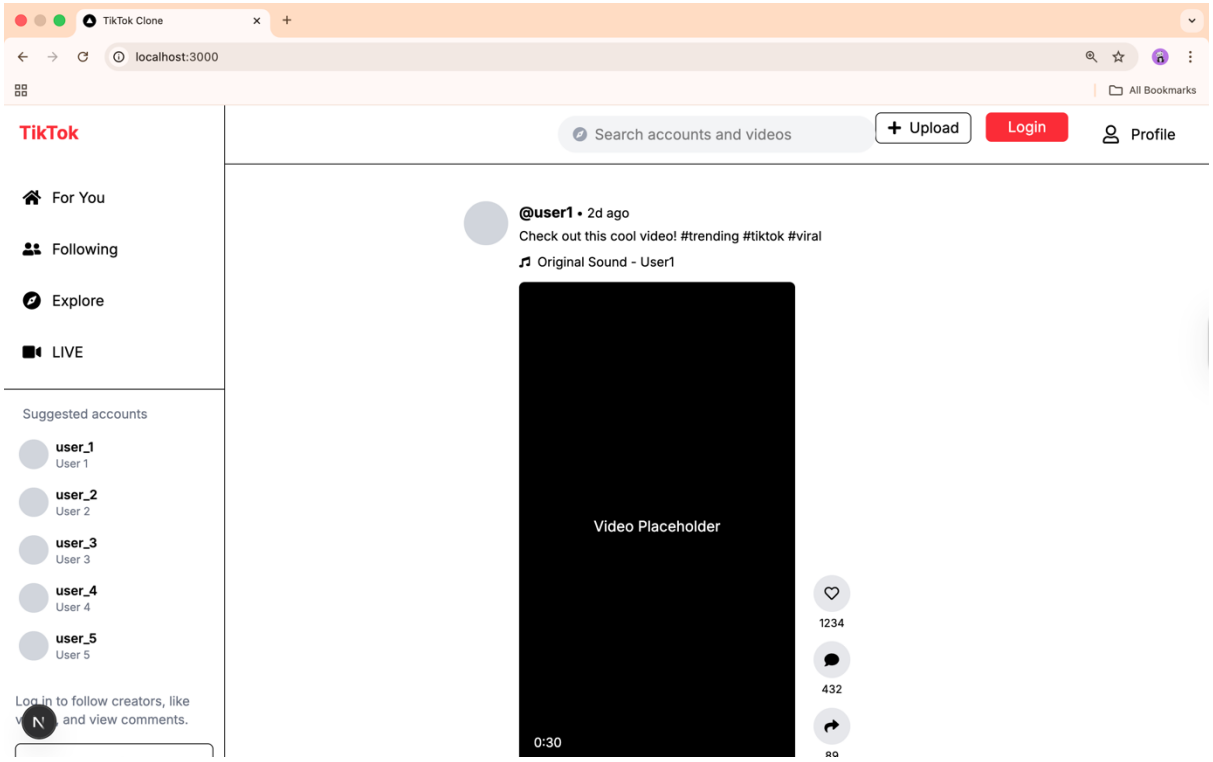
Step 1: Install additional dependencies using: npm install react-icons

Step 2: Update MainLayout.jsx with a full TikTok-style sidebar navigation including icons.

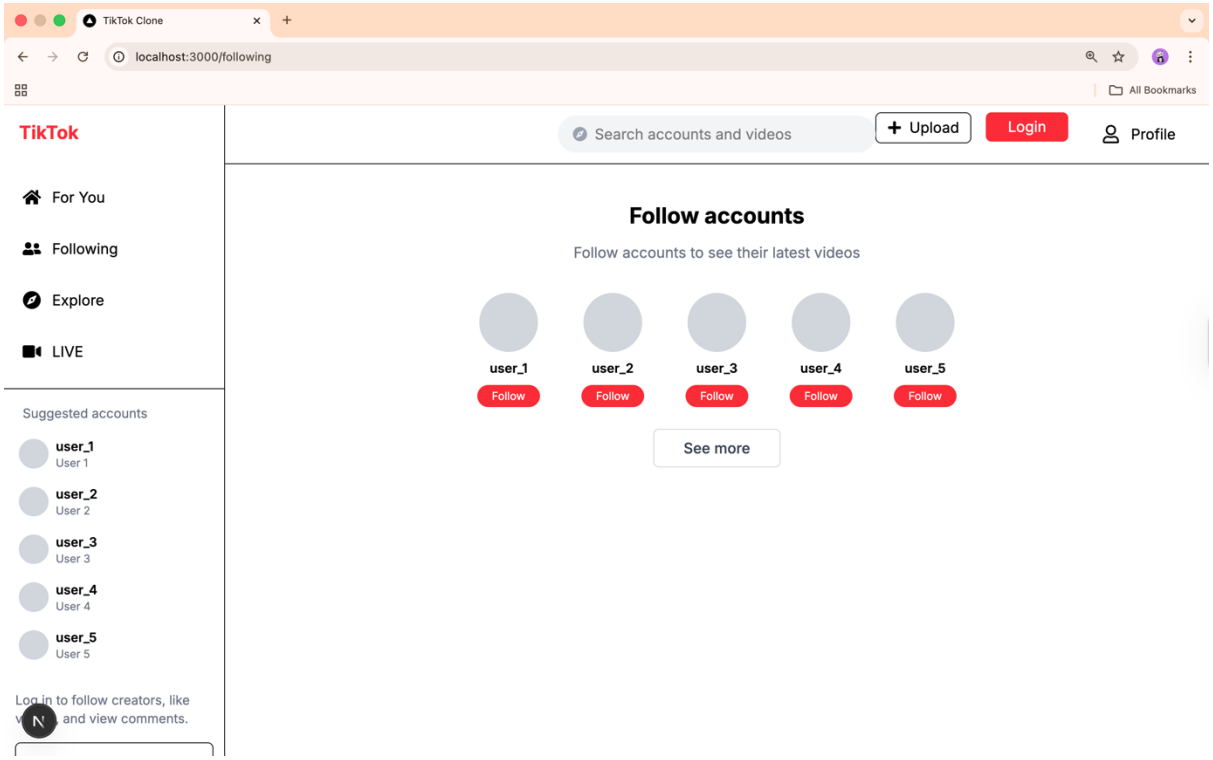
Step 3: Create the VideoCard component at src/components/ui/VideoCard.jsx with like, comment and share buttons.

Step 4: Create the VideoFeed component at src/components/ui/VideoFeed.jsx with sample video data.

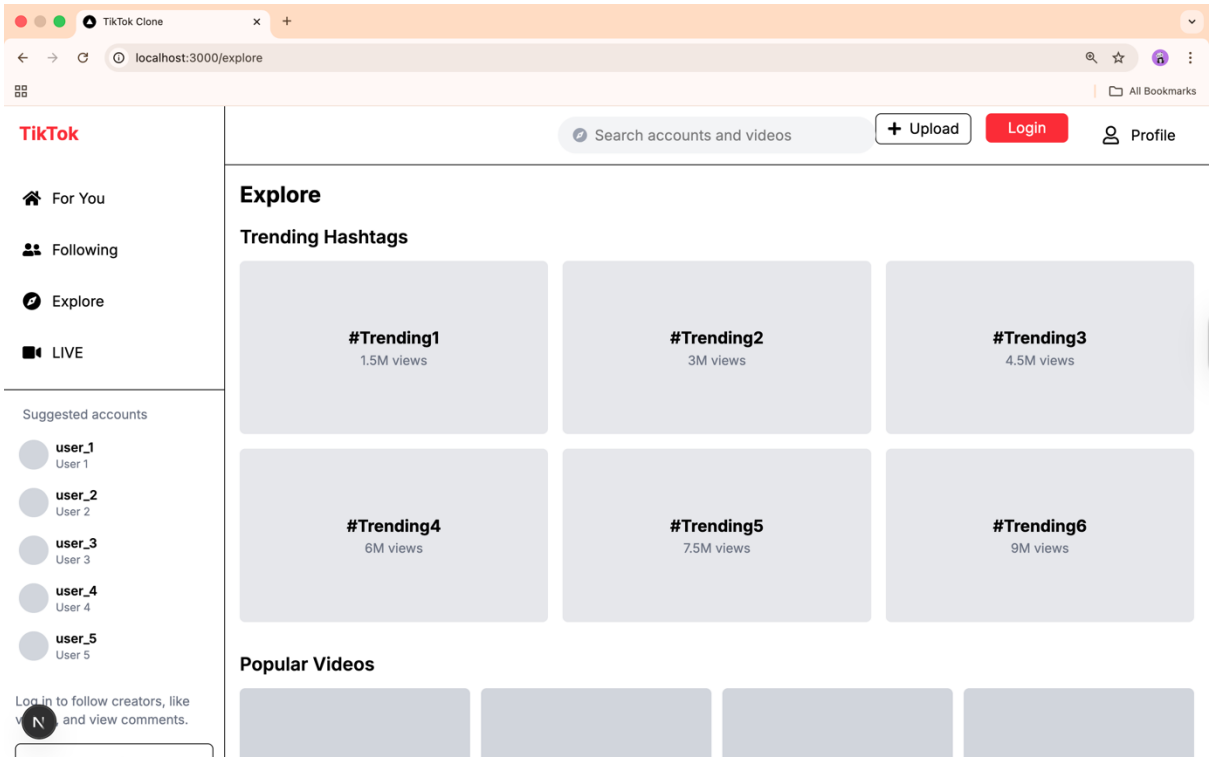
Step 5: Update the home page src/app/page.js to display the VideoFeed component.



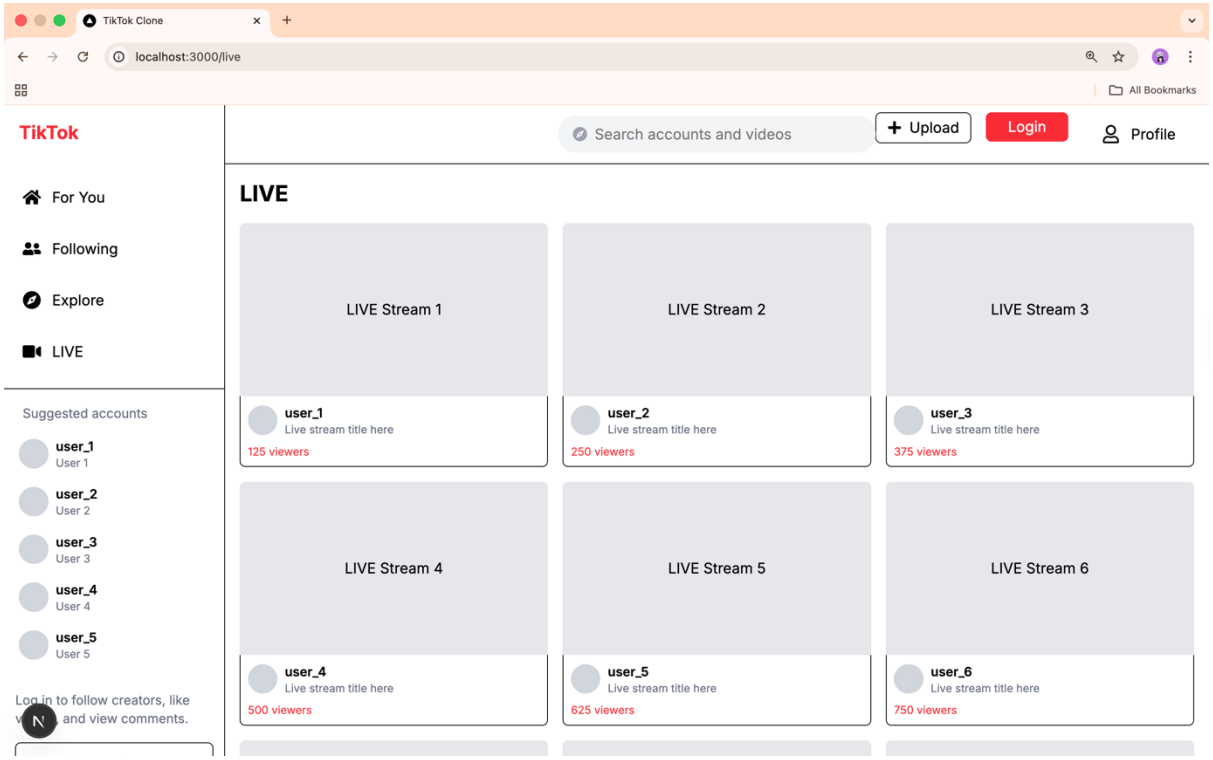
Step 6: Create a Following page at src/app/following/page.jsx.



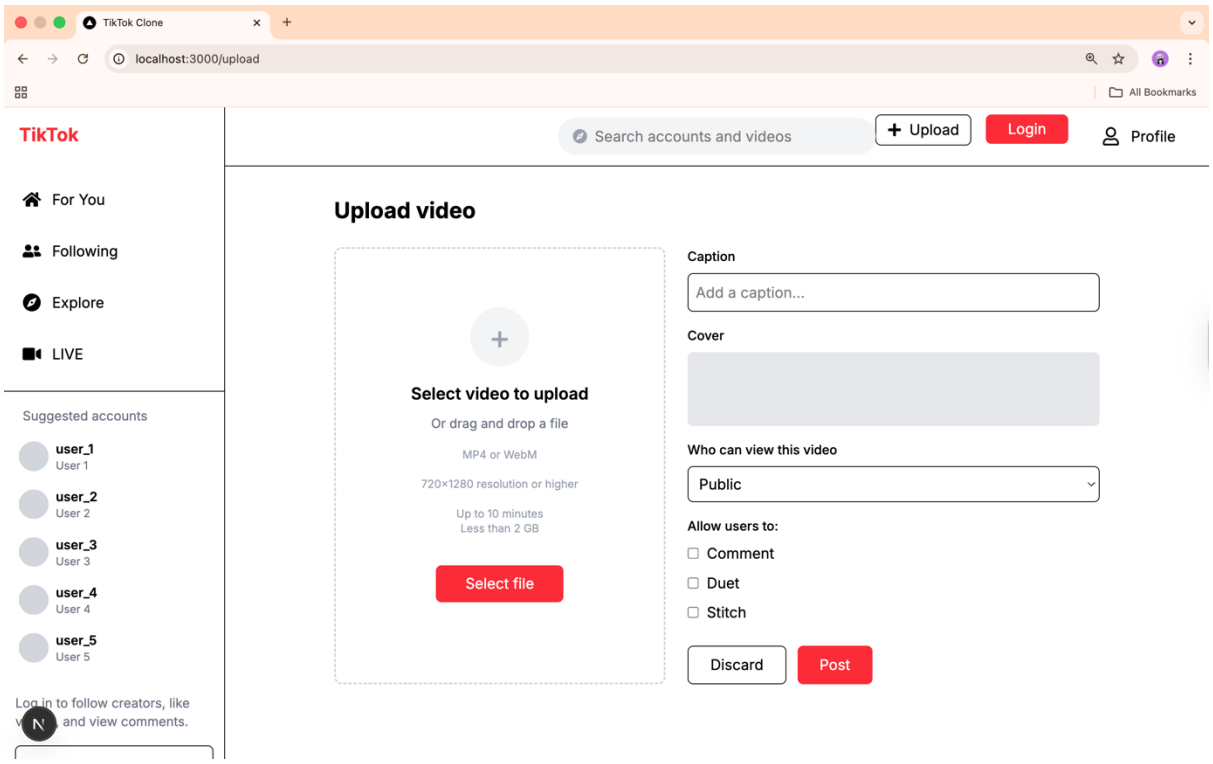
Step 7: Create an Explore page at src/app/explore/page.jsx.



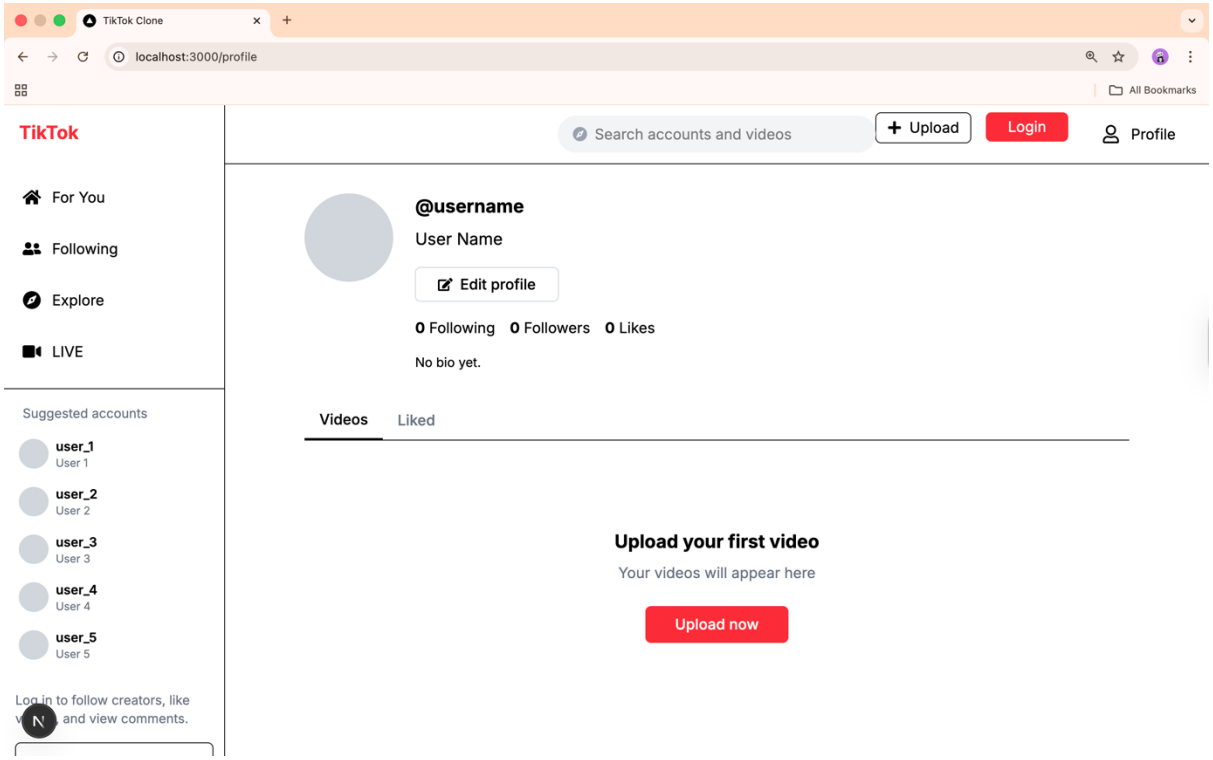
Step 8: Create a Live page at src/app/live/page.jsx.



Step 9: Update the Upload page at src/app/upload/page.jsx.



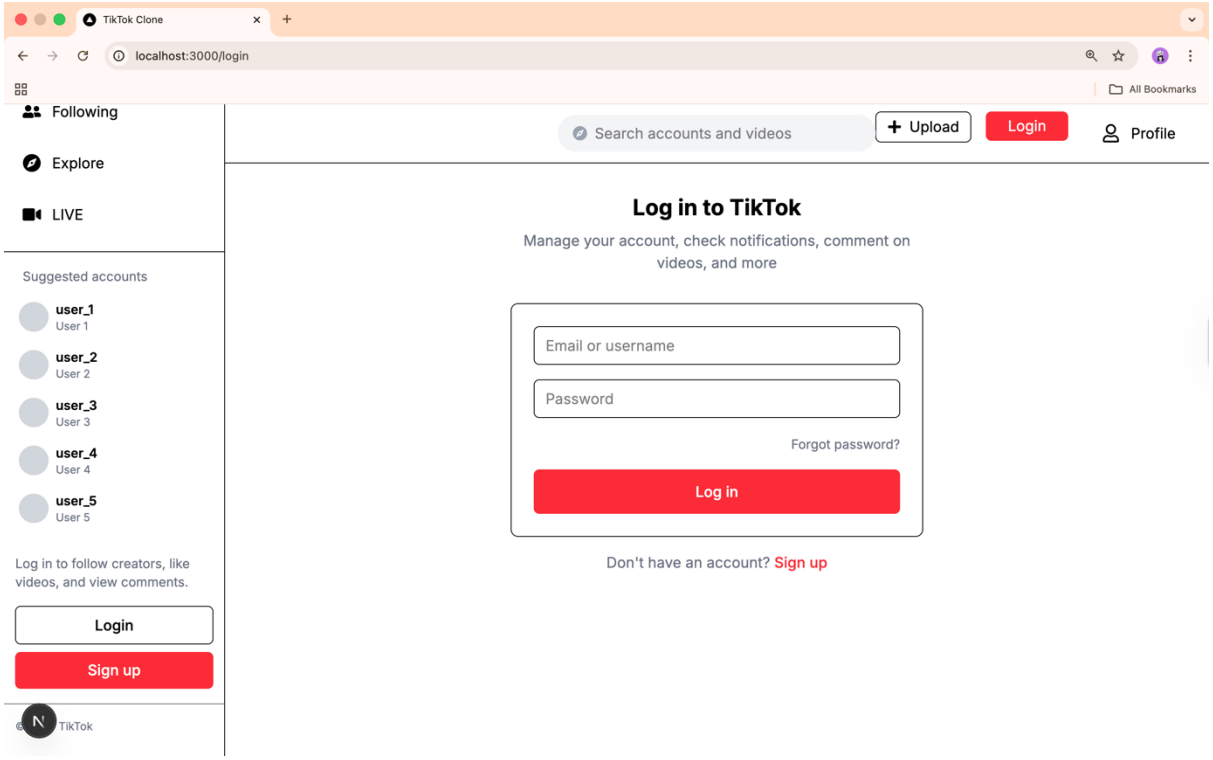
Step 10: Update the Profile page at src/app/profile/page.jsx.



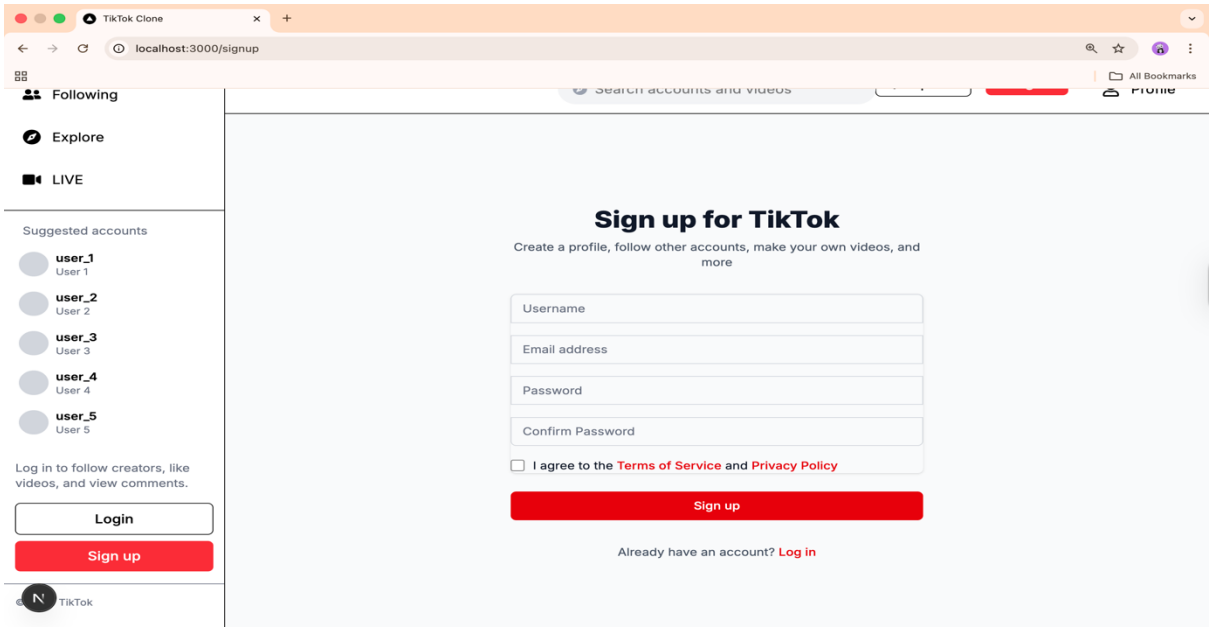
Part 3 - Creating Login and Registration Forms

Step 1: Install required dependencies using: npm install react-hook-form

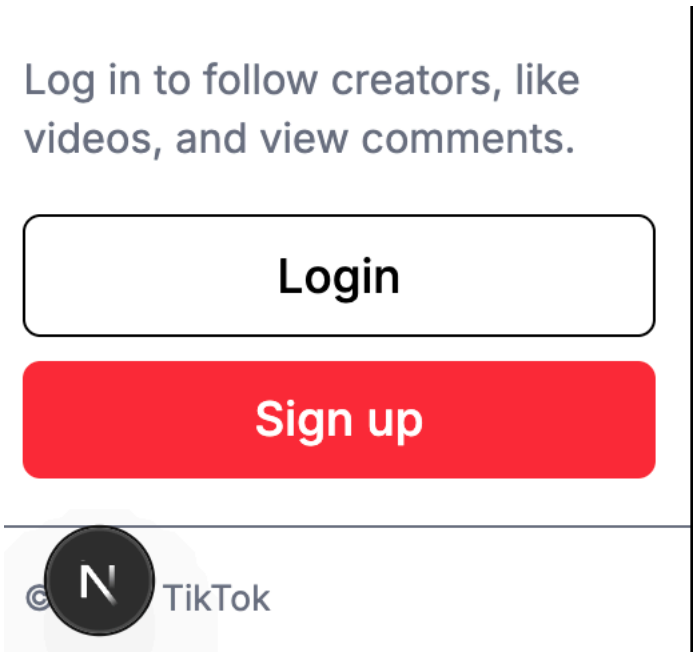
Step 2: Create a Login page at src/app/login/page.jsx with email and password fields and form validation.



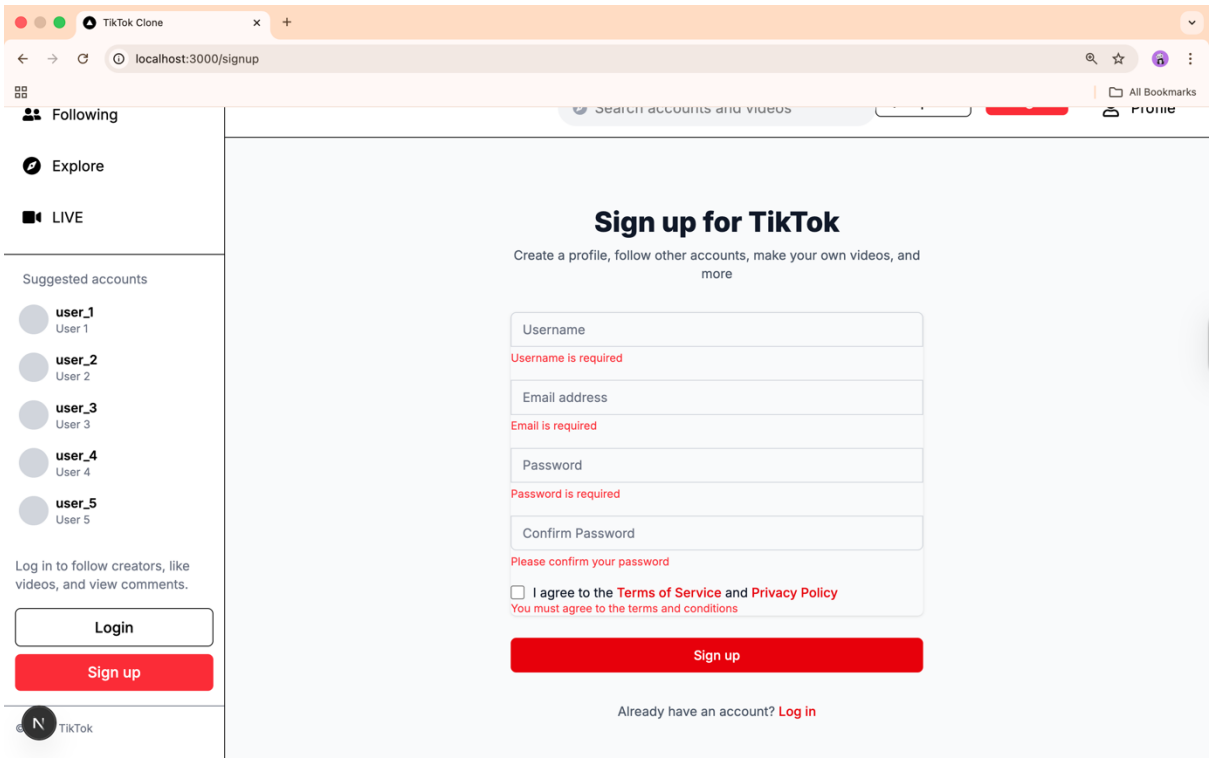
Step 3: Create a Signup page at src/app/signup/page.jsx with username, email, password, confirm password and terms checkbox fields.

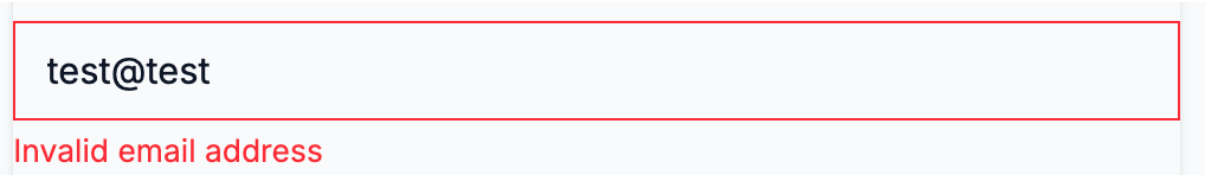


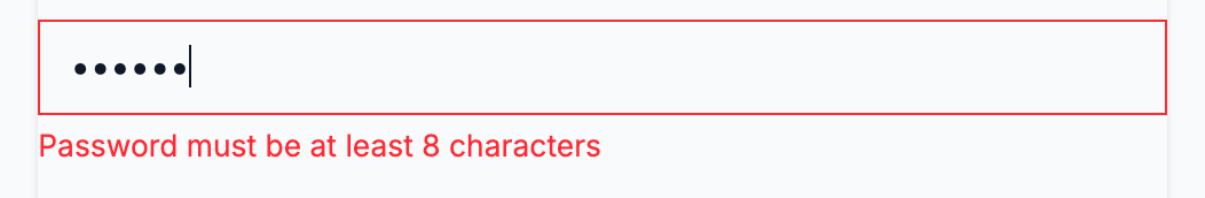
Step 4: Update MainLayout.jsx to connect the Login and Signup buttons to the correct pages.

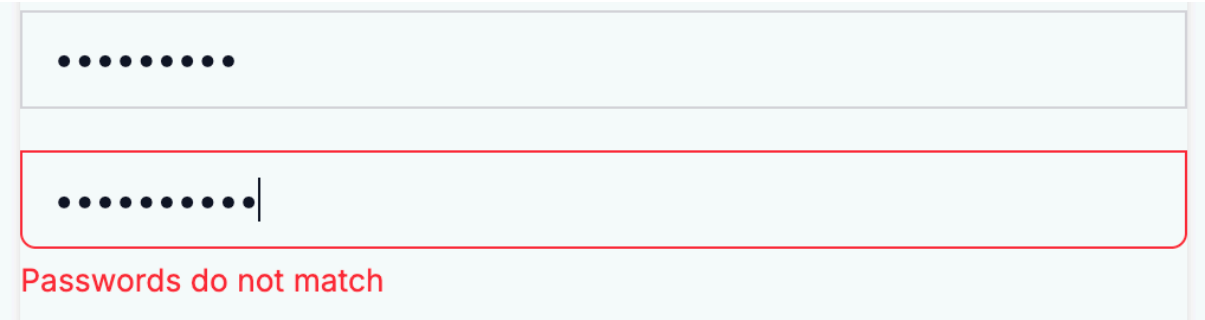


Step 5: Test form validation:

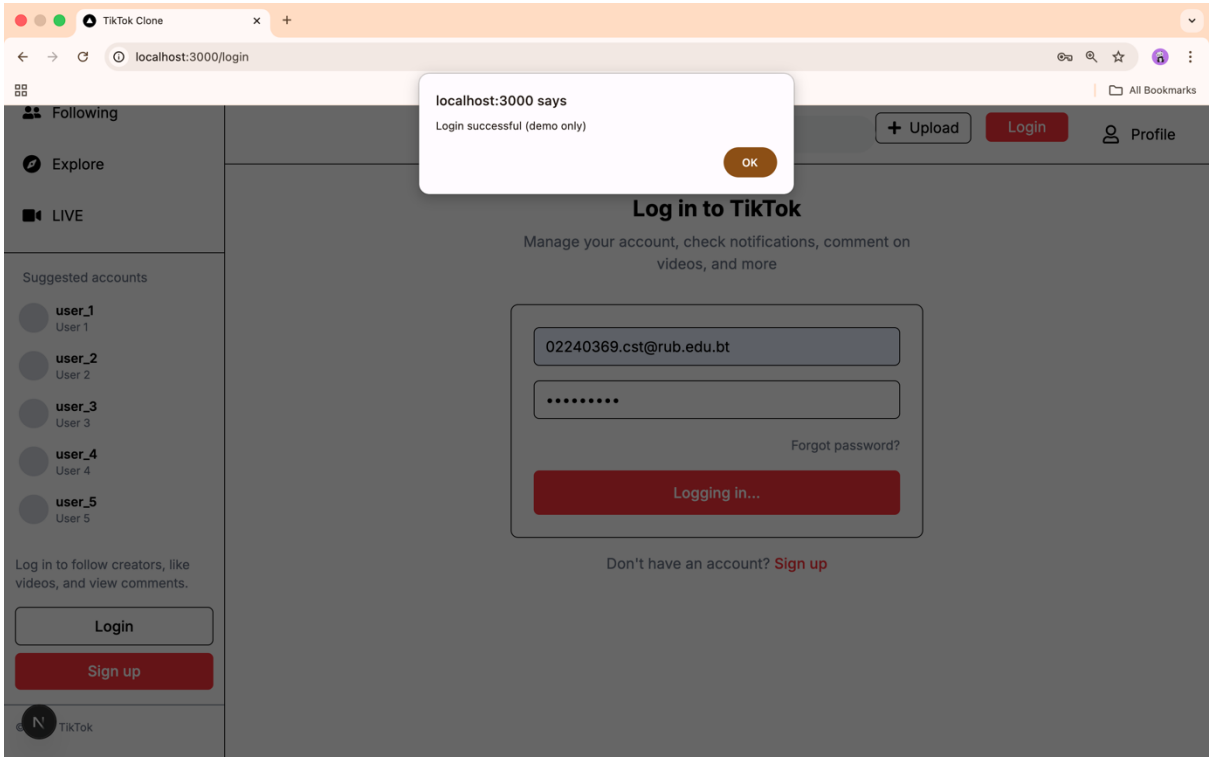


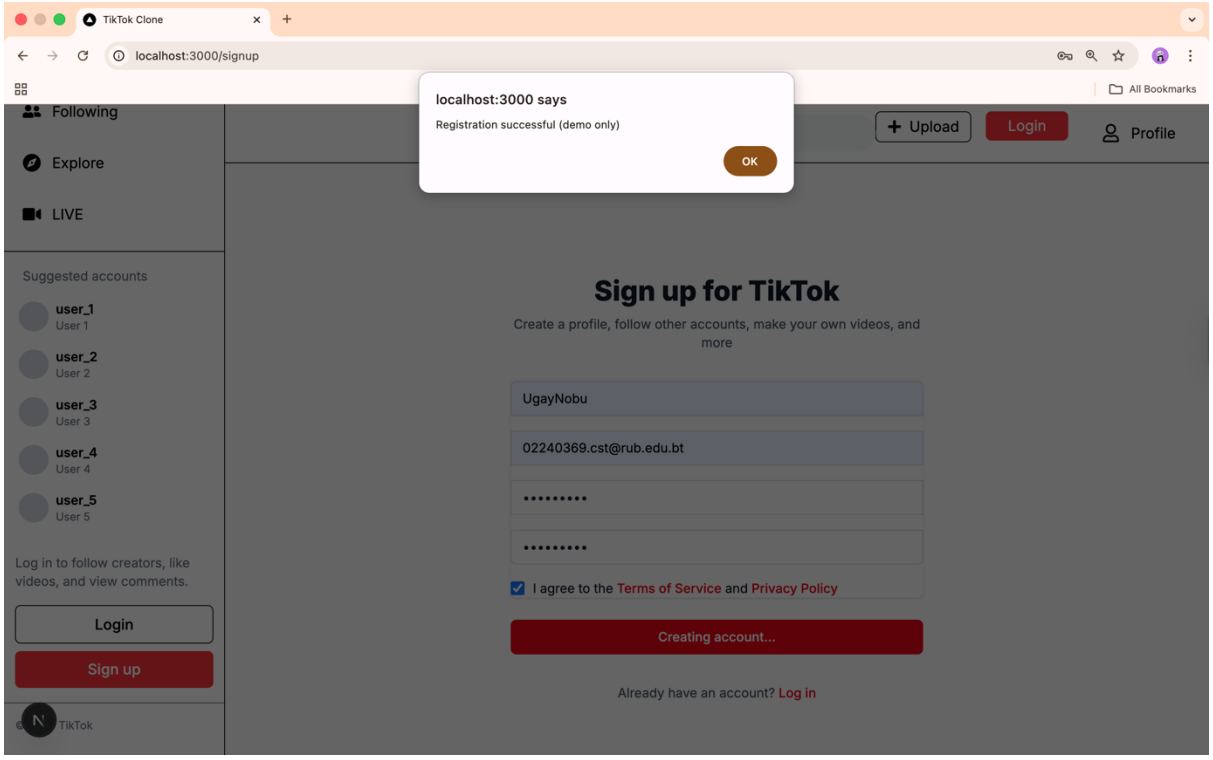






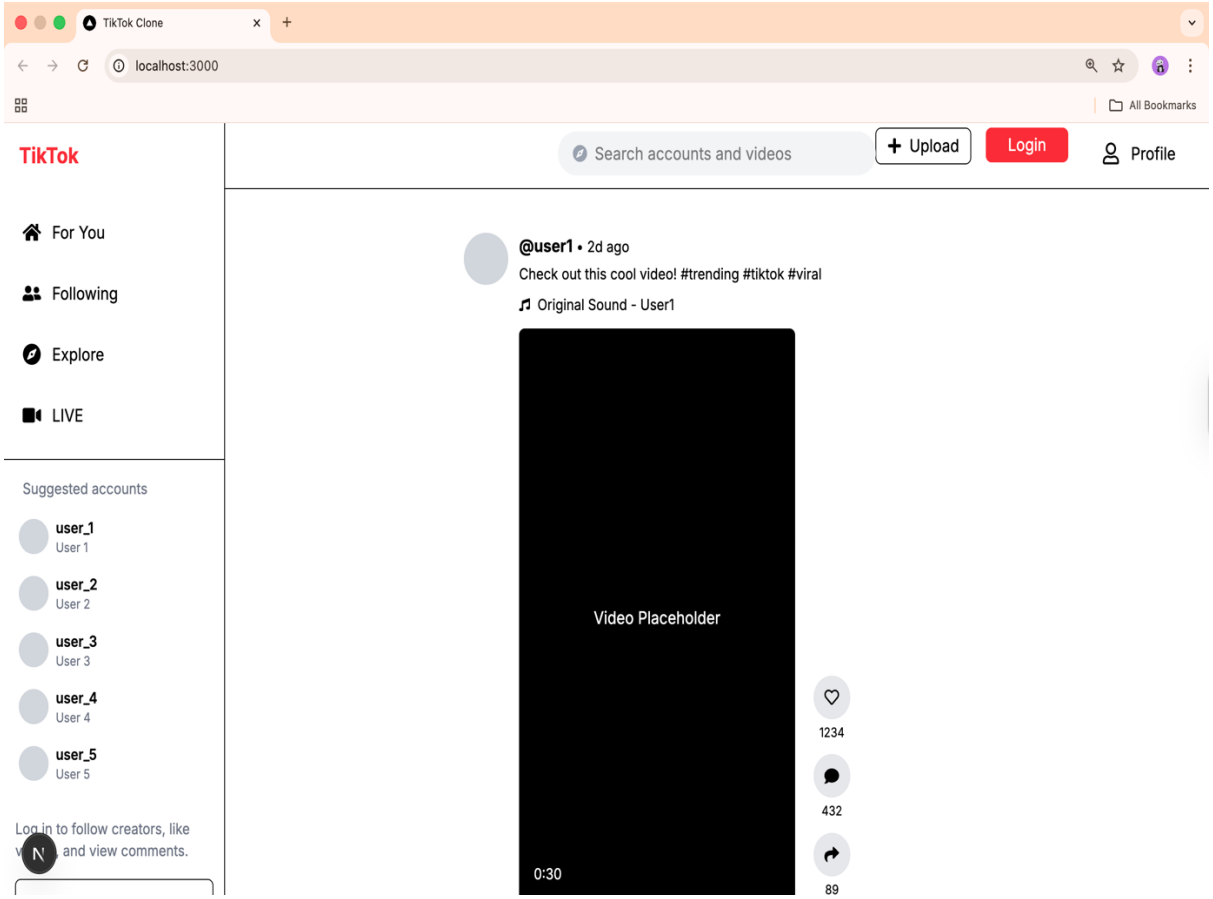
Step 6: Test successful submission:

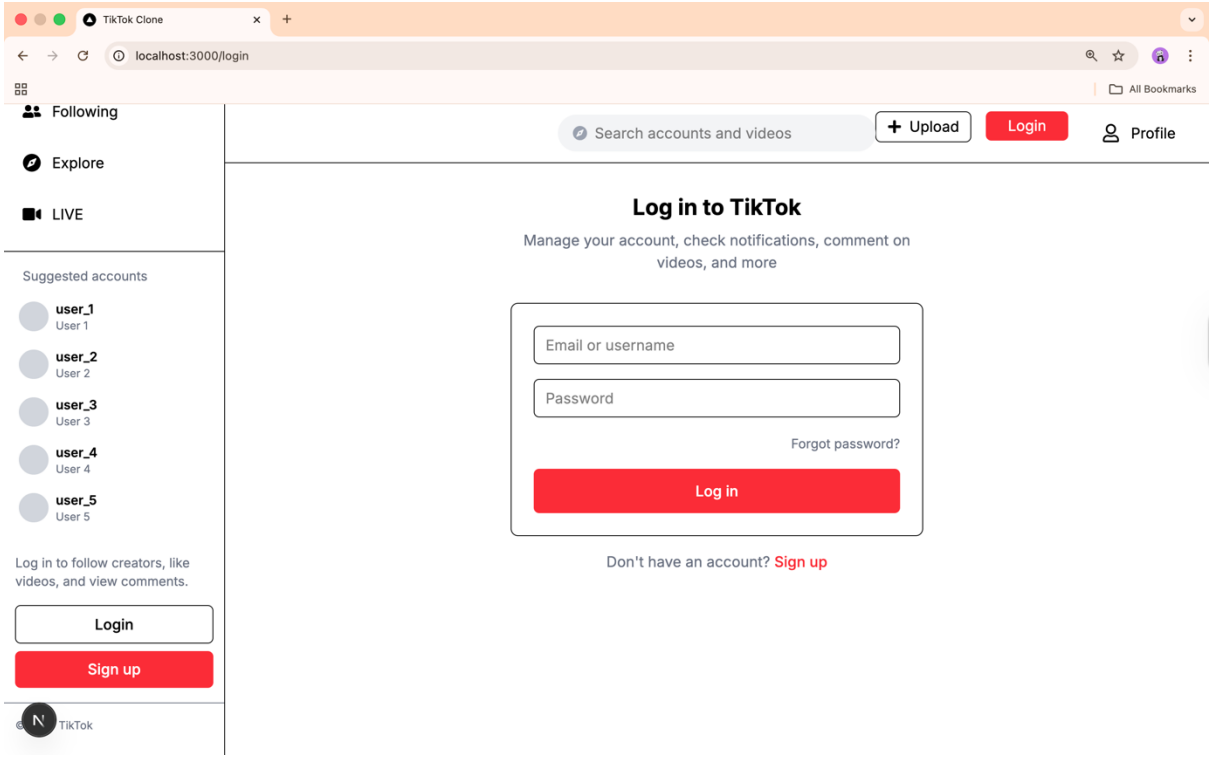


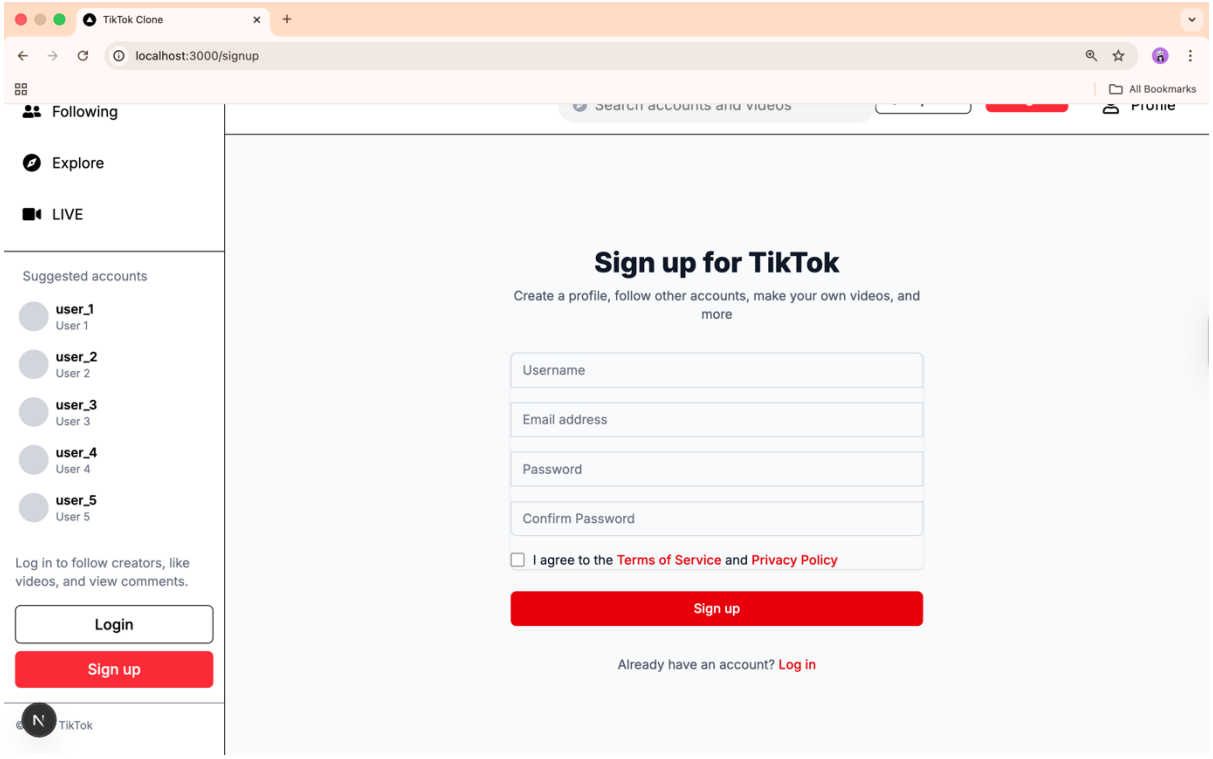

TikTok Clone
localhost:3000/signup
All Bookmarks
localhost:3000 says
Registration successful (demo only)
OK
Following
Explore
LIVE
Suggested accounts
user_1
User 1
user_2
User 2
user_3
User 3
user_4
User 4
user_5
User 5
Log in to follow creators, like videos, and view comments.
Login
Sign up
TikTok
Upload
Login
Profile
Sign up for TikTok
Create a profile, follow other accounts, make your own videos, and more
UgayNobu
02240369.cst@rub.edu.bt
I agree to the Terms of Service and Privacy Policy
Creating account...
Already have an account? Log in

Final Output

This is where I show my talent by showing the final fully workable TikTok functions made using the Next.js and Tailwind CSS. It has all the necessary requirements like the navigable sidebar with proper pagination, for you vudeo feeds, many different pages like the upload and profile, and mainly working valid login and signup forms with validation.




TikTok Clone
localhost:3000/login
All Bookmarks
Following
Explore
LIVE
Search accounts and videos
Upload
Login
Profile
Log in to TikTok
Manage your account, check notifications, comment on videos, and more
Email or username
Password
Forgot password?
Log in
Don't have an account? Sign up
Suggested accounts
user_1
User 1
user_2
User 2
user_3
User 3
user_4
User 4
user_5
User 5
Log in to follow creators, like videos, and view comments.
Login
Sign up
TikTok


TikTok Clone
localhost:3000/signup
All Bookmarks
Following
Explore
LIVE
Sign up for TikTok
Create a profile, follow other accounts, make your own videos, and more
Username
Email address
Password
Confirm Password
I agree to the Terms of Service and Privacy Policy
Sign up
Already have an account? Log in
Suggested accounts
user_1
User 1
user_2
User 2
user_3
User 3
user_4
User 4
user_5
User 5
Log in to follow creators, like videos, and view comments.
Login
Sign up
TikTok

CONCLUSION

Finally, after a lot of errors and syntax I was finally able to successfully make a workable TikTok-based web page, even if it is just locally by using frameworks like Next.js, Tailwind CSS and library React Hook Form. This project is divided in 3 major. Part 1, where I set up the Next.js project folders, styling using the Tailwind CSS and make basic folder structure with all the necessary files that act as pages in the application. Part 2, here I made the main layout with a permanent navigable sidebar, made VidooCard and VideoFeed files for you page and add many different pages such as profile, upload, etc... In part 3, I focused on making the proper register forms like login and sign up with standard validation by using the React Hook Form Library.

I faced a lot of problems or issues like apostrophe syntax in the JSX files, I had many issues where the working navigation paths didn't work properly and had had time completely understaning the foder sturecure where I didn't know to cd into the proper folders to install the nessary dependences for the particular features to apply in the application since all of this was my first time working as a website developers that to only frontend. However, I was getting used to it over time with more practice and help from my tutors where I had to carefully use my brains to analyse each error messeges and make the required edits or corrections.

This assignment has given me the chance to build a strong foundation by understanding basic developments in the new era of web application using React and Next.js, which is my first time dealing with these devices where I found the component-based approach was the best to arrange or organize and reuse of codes. This new knowledge will definitely be useful down the line in my career path or soon to come assignments, and I will be looking forward to it in the future.

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